Project Overview

CodeBotix

Presented By:
Sigma

Agenda:

/01 Key Insights

/02 Highlights & Suggestion from Students

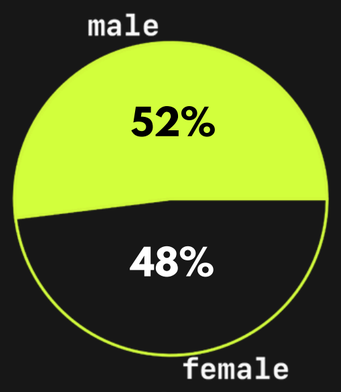
/03 Survey Classification

/04 Suggestive Feedback Questions

Key Insights

**All Data are Mean Values

A healthy Sex-Ratio is demonstrated, as targeted by the Client, with equal participation and equal interest.

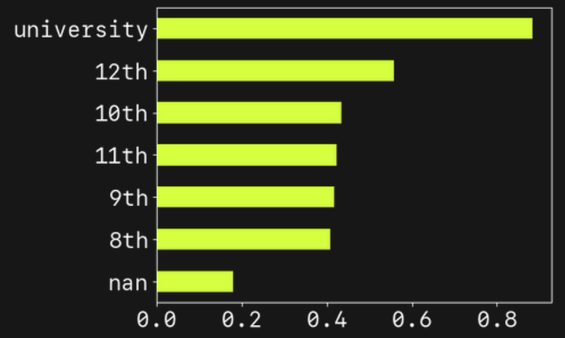


92.6% attendees have shown an increase in interest in Robotics after the workshop. Interest of the attendees in Robotics has seen a **1.5X** increase on average, after the workshop

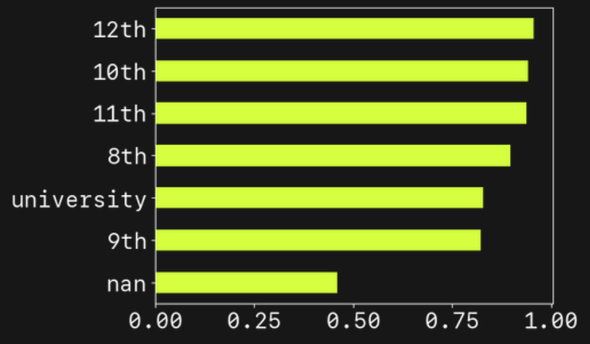
Interest Change:

Interest in Robotics **before** the workshop increases with education level, peaking among University students. However, interest dropped for University students **after** indicating lack of curriculum for University students.

Interest Before:



Interest After:

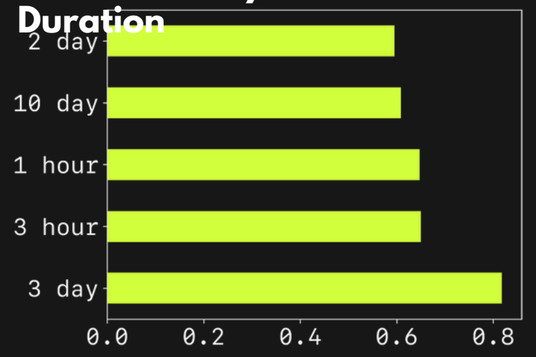


Key Insights

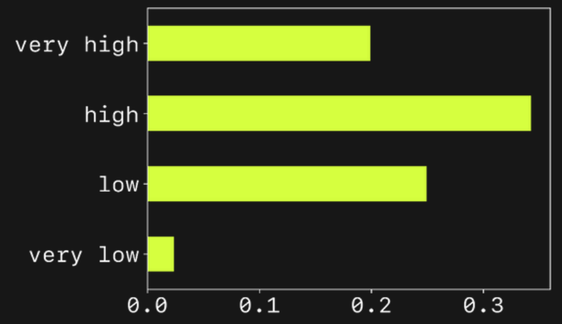
**All Data are Mean

The 3-day workshop is the most **successful** in terms of attendee Satisfaction. Students prefer **1hr, 3hr** for shorter sessions and for lengthy workshop a sweet spot would be **3-day.** Even Knowledge gain was **higher** in **3-day.**

Satisfaction By Duration



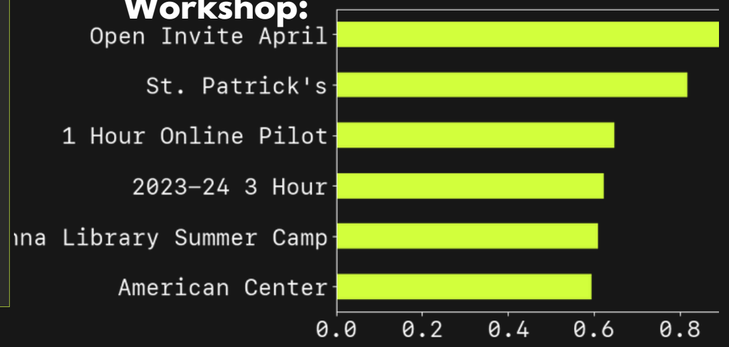
Interest Change By Experience



Students who already had **prior** experience and knowledge showed more **positive** interest towards the course after the Workshop. Indicating potential students with high **retainability** and **satisfaction**

Open Sessions have **higher** expectations maybe because the **less** depth of content. Also, Students showed equal interest on both online/offline medium.

Satisfaction By Workshop:



Making of the database

How were the numbers obtained?

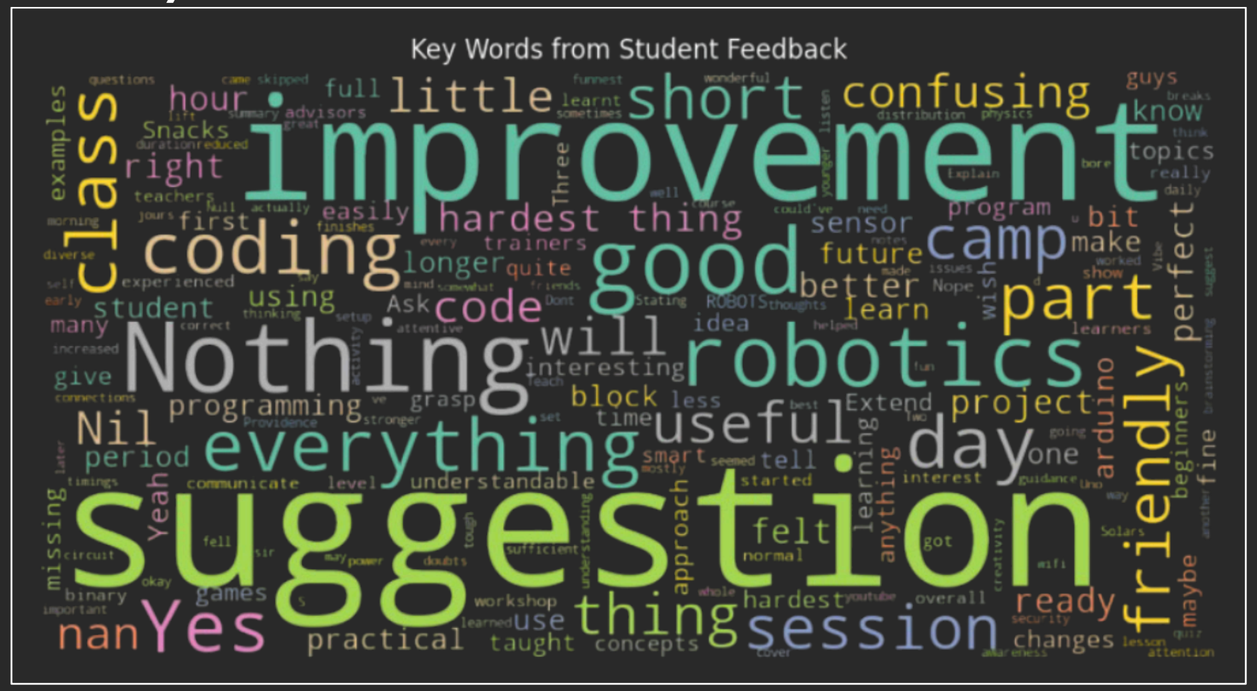
Though there were about 80 unique questions, they all helped understand only about 13 important characteristics of the participants:

1. name
2. gender
3. class
4. mode
5. workshop_name
6. workshop_duration
7. interest_before (interpreted)
8. interest_after (interpreted)
9. highlights
10. Suggestions
11. Satisfaction (interpreted)
12. Expectations
13. prior_experience (interpreted)
14. knowledge_gain (interpreted)

The interpreted columns like interest_before, knowledge gained etc. have numerical values. Eg: 0.985. This number on the 'interest' of the person is in a lose sense, the probability that the person is interested in Robotics based on his answers to various questions that gauge interest. The number was derived by zero-shot-classification using the Facebook BART NLP model locally.

Why Zero Shot Classification?

- Helps compare both numerical and textual answers.
- Sensitive to tone of the textual answers
- Easy to track change in quantities like 'interest', 'satisfaction'.
- Gives a simple probability like quantity perfect for analysis.


Key Words from Student Feedback
improvement
suggestion
Nothing
everything
robotics
coding
class
good
camp
part
friendly
day
useful
Yes
nan
session
thing
hardest thing
confusing
short
little
hour
Snacks
right
first
easily
longer
trainers
code
Ask
using
student
many
give
Nil
period
programming
grasp
block
interesting
will
idea
learn
better
future
sensor
program
make
bit
perfect
know
guys
topics
examples
Three
full
advisors
learnt
sometimes
distribution
physics
explain
really
daily
worked
show
Nope
made
issues
learners
beginners
fine
arduino
project
Extend
one
interest
guidance
started
tell
smart
seemed
mostly
time
less
best
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going
Providence
stronger
set
Teach
connections
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learned
games
binary
important
okay
fell
sir
power
doubts
sufficient
understanding
tough
circuit
Solar
creativity
wifi
youtube
security
lesson
quiz
awareness
attention
cover
whole
another
brainstorming
suggest
Vibe
worked
made
issues
show
wish
Nope
stating
ROBOTS
thoughts
helped
learners
fun
Uno
way
guidance
going
interest
started
seemed
mostly
time
less
best
two

Suggestions From Students

Extend the duration / sessions are too short:

- Many felt 2 hours/day or 3–10 days was not enough.
- Requests to extend sessions or add more days.

More practical / hands-on activities:

- Multiple participants wanted more practical robotics work instead of just theory.

Coding was difficult / needs more explanation:

- Students found programming/coding parts tough, asked for clearer explanation and more time.

More breaks / better timing:

- Some suggested longer morning sessions or additional breaks.

Some topics skipped / want more coverage:

- Especially sensors, electronics, and advanced programming.

Highlights:

What Students appreciated the most?

1. Coding & Programming:
   - Block coding
   - Data types & binary code
   - General coding concepts

1. Arduino & Sensors:
   - Using Arduino
   - Ultrasonic sensors
   - Combining hardware with programming

1. General Robotics
   - Basics of robotics
   - Interior of robots
   - Normal level robotics concepts
   - Laws of robotics
   - Hands-on use of Arduino with sensors

Survey Classification

Based on the data, we understand that our client offers a variety of sessions and workshops. We will categorize the survey responses and questions according to these requirements, ensuring that future analysis remains efficient, clear, and actionable.

Surve
y:

For Small Sessions(Online, 1hr, 3hr, 1 day):

Conduct only 2 Surveys:

- **During Registration:** Provide personal information, academic details, parent info, age, school, grade, interests, experience and expectations.
- **Feedback:** Feedback from students and parents on course content, kit usability, teaching quality, competitions, what they learnt, takeaways and suggestions of Improvements

For Open Sessions:

Conduct only 1 Surveys:

- **Feedback:** Stick to only a feedback form for suggestions, likeness and whether they will enroll in future again.

For Bootcamps/ Long Workshop (10 days, 3 days):

Conduct only 3 Surveys:

- During Registration
- In-between Session: difficulty, teacher review, suggestions, learning curve, how engaging the sessions are, is the pace alright.
- Feedback

Form 1: Easy for EDA

From the analysis, we understood that most of the questions should be structured as **MCQs** rather than text-heavy responses. This approach offers three key benefits: **reduced effort** for students to complete the forms, **simplified** data analysis, and improved **consistency** in responses.

Survey Framework:

- The survey will now contain majorly only questions with **numerical** answers—answers that are either numbers themselves which can be mapped to a number.
- Options should be given for text-based questions.
- All workshops should have the **same** questions with same **wording**. (Reuse the same questions—even if not all of them)
- All workshops should have 2-3 surveys depending on workshop duration.

G-Form Link

Examples:

Pre-workshop Survey

How interested are you in robotics? *

- Way too much (5)
- A lot (4)
- Interested (3)
- Not much (2)
- Not at all (1)

How much do you already know about robotics? *

- Way too much (5)
- A lot (4)
- A decent amount (3)
- Not much (2)
- Not at all (1)

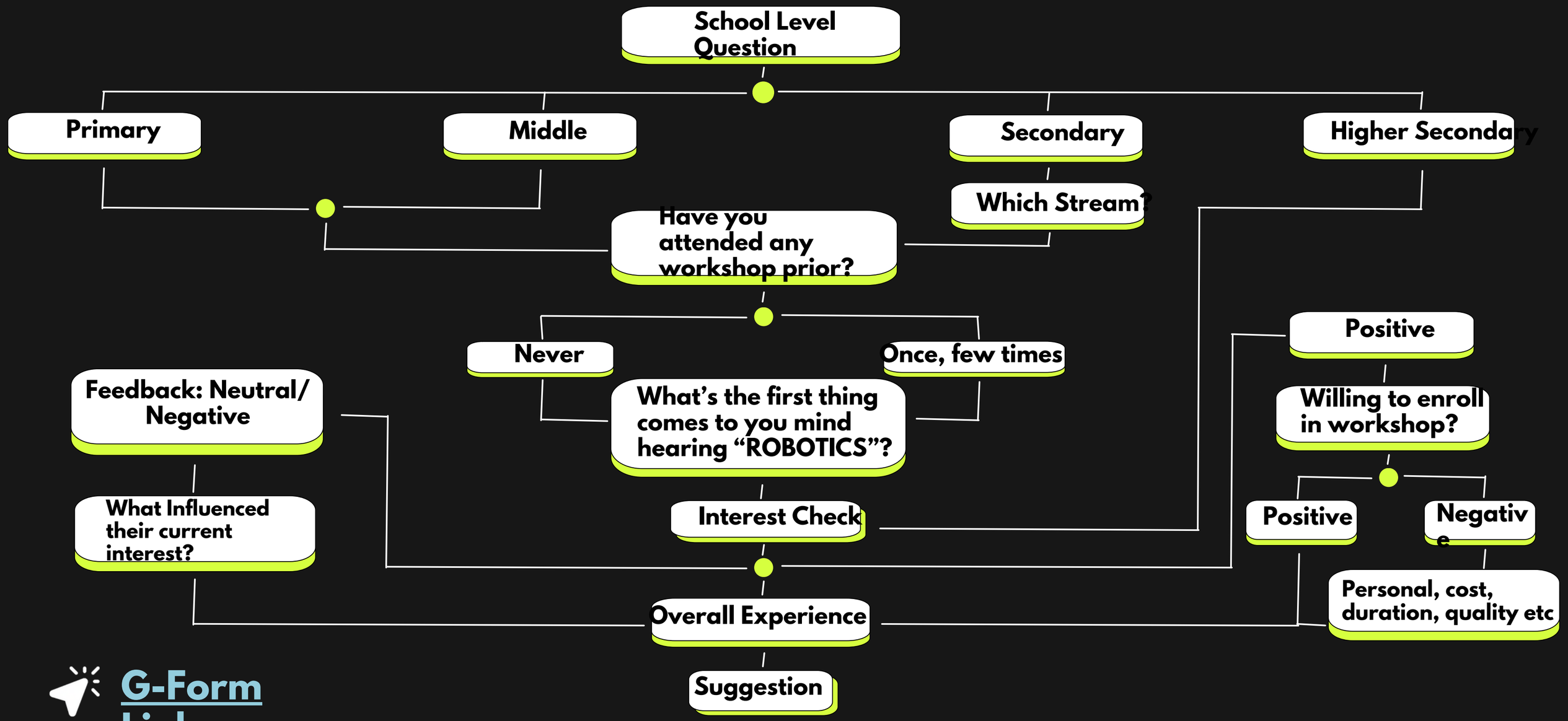
What do you expect from the workshop? *

- [ ] Introduction to Robotics (Theory)
- [ ] Interesting live demonstrations
- [ ] Practical experience in making a robot
- [ ] Guidance in pursuing robotics as a career path
- [ ] Other:

How satisfied are you with the workshop? *

- I had the best time ever! (5)
- It was very fun (4)
- It was good enough (3)
- It was not that good (2)
- It was bad (1)

Form 2: Easy for Survey Taker



G-Form Link

Feedback Question Bank

1. Interest Before

Q. How interested are you in learning robotics through this workshop?

2. Prior Experience

Q. Have you attended a robotics workshop or worked with circuits/coding /microcontrollers before?

3. Highlight Factor

Q. What was the most valuable thing you learned in this workshop?

4. Satisfaction

Q. How satisfied are you with this workshop overall?

5. Trainer Rating

Q. How would you rate the trainers?

6. Interest After

Q. After attending this workshop, how interested are you in pursuing robotics further?

7. Suggestions to Improve

Q. What should we improve in future workshops?

8. To Improve (Content/Setup)

Q. Were any topics/tools missing or unclear in this camp?

9. Expectations

Q. Were your expectations from this workshop met?

10. Knowledge Gain

Q. How would you rate your understanding of robotics after this workshop?

Suggestive Solution

Phase 1: Immediate action

- Feedback and Survey form: Roll out new forms with better and targeted questions. Every Workshop should survey and feedback for future improvements.
- The gives an overview how G-forms should be.

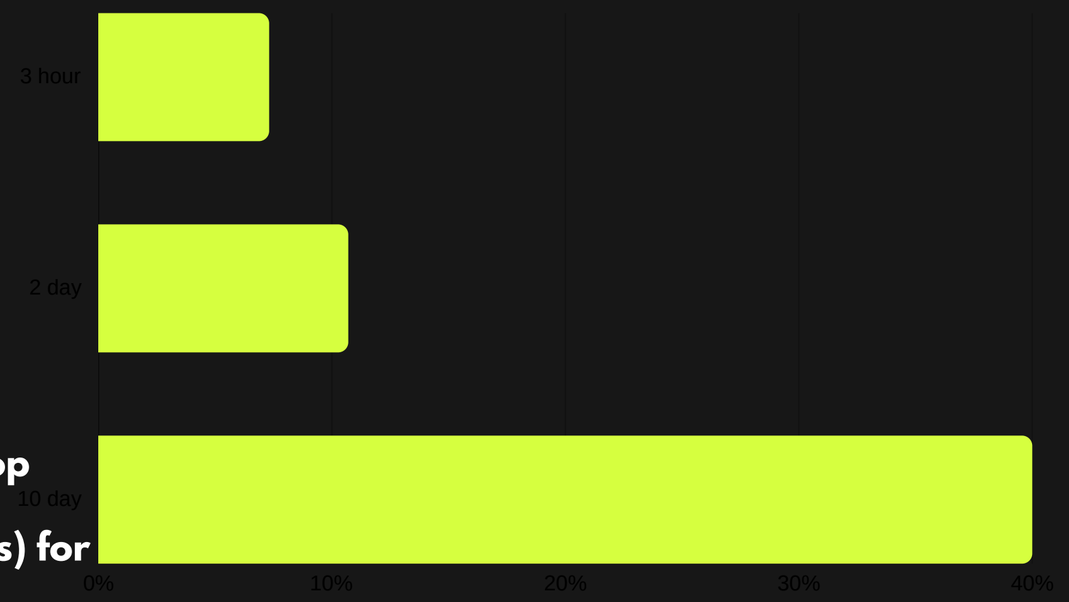
Phase 2: Addressing Current Issues:

- Based on Current Feedback and data we got, the key suggestions from Students:
  1. Optimize duration for Each Workshop: Divide the entire workshop into focused mini-modules, each with a clear outcome.
  2. Too much theory: Share short pre-session videos/notes (10–15 min) covering theory. Use live time only for hands-on tasks and troubleshooting.
  3. Introduce Breaks with Refreshments: will act as hookups.
  4. Gamification and competitions: Integrate friendly contests, badges, and leaderboards tied to actual skill milestones.

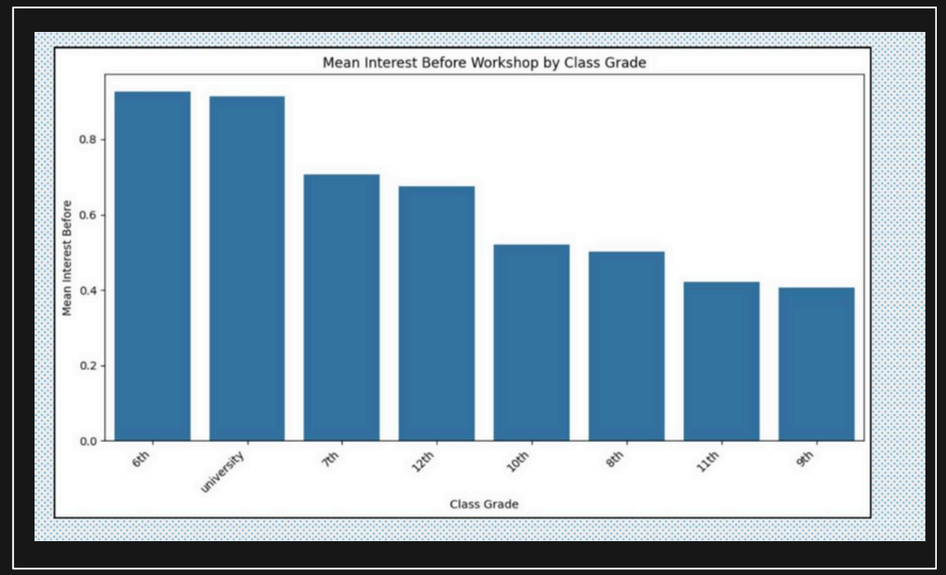
Phase 3: Retention Strategy:

1. Program: “Codebotix Achievers Loop”:
   1. Award digital badges, certificates, and social media shoutouts for top performers.
   2. Offer small incentives (discounts / early access / special kit upgrades) for students who re-enroll or refer friends.

Module	Duration	Focus
Module 1	1 day	Basics of Robotics + Sensors Demo
Module 2	1 day	Arduino + Simple Projects
Module 3	1 day	Coding Concepts + Hands-on Task
Module 4	½ day	Integration Project + Presentation



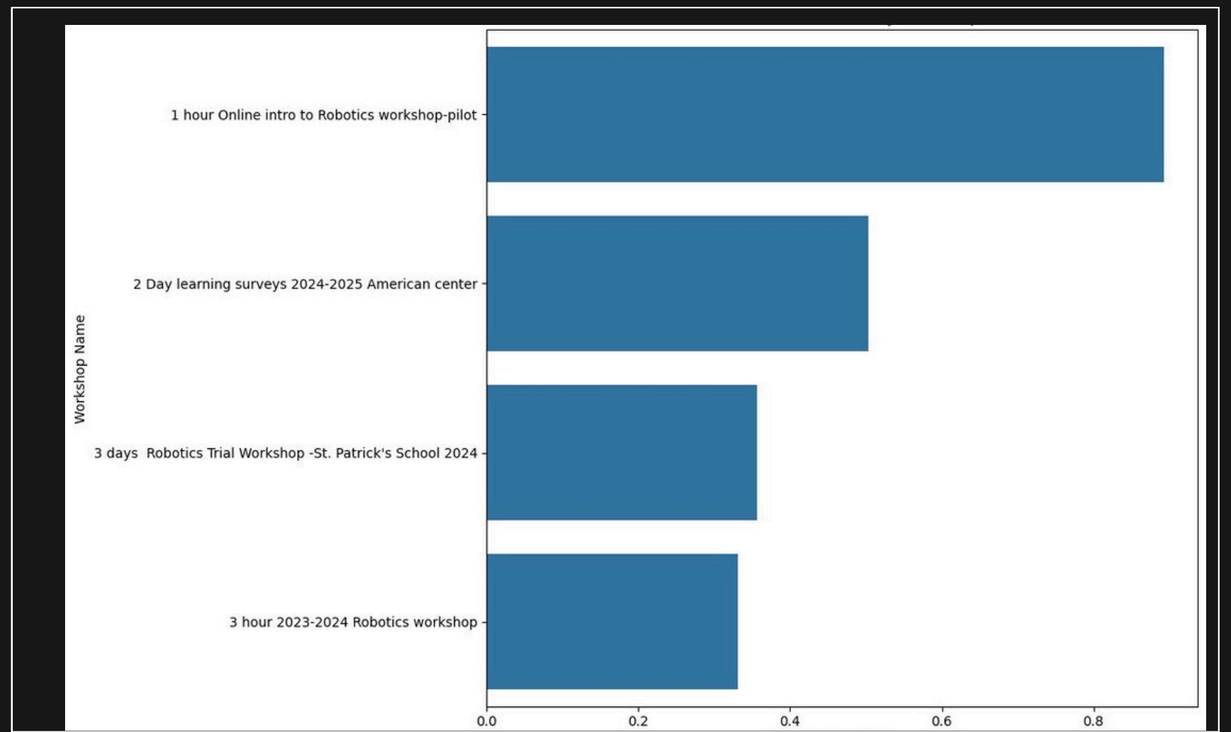
DATA ANALYTICS



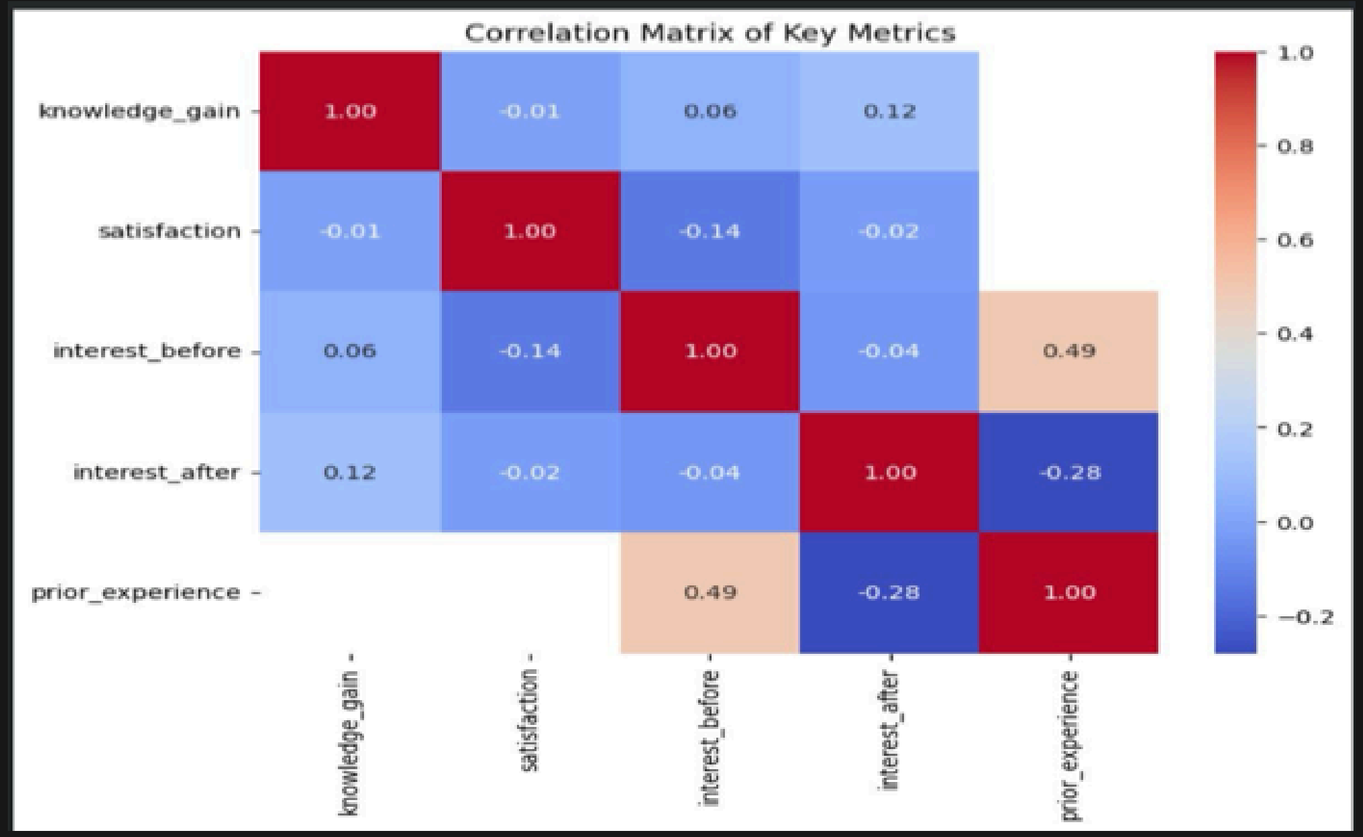
Mean interest before workshop by class grade

- **Students in lower grade(6th and 7th) and university students reported higher initial interest** in workshops than other students.
- This could be due to a greater sense of novelty and curiosity about STEM topics in younger students and and a more focused interest in specialized fields among university students

Mean interest by workshop

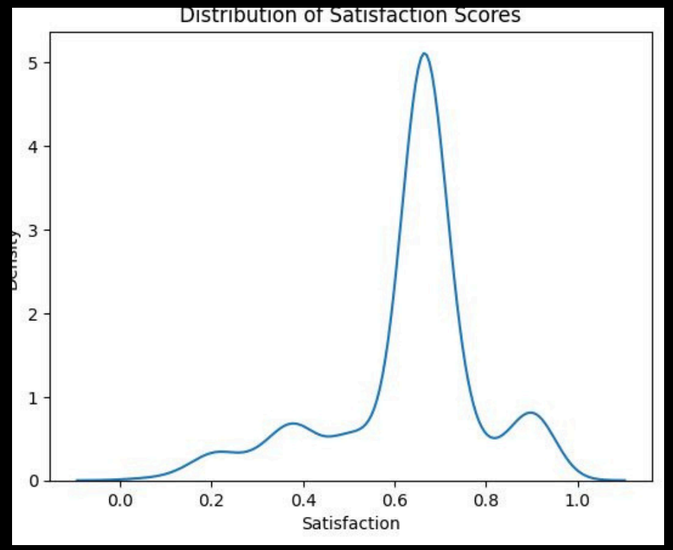


Correlation Matrix

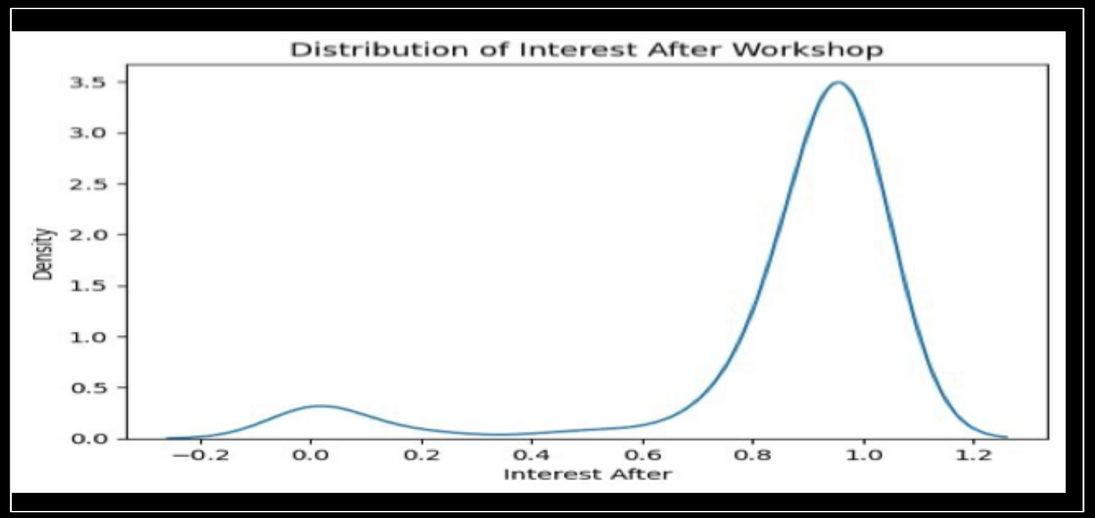


DATA ANALYTICS

Distribution of satisfaction scores



Distribution of Interest after workshop



THANK YOU

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